Thermistors are rugged in construction, cheaper in cost and highly sensitive devices.

h. Advantages of resistance transducers

i. Simple in construction
ii. High output
iii. Less expensive
iv. Available in different shapes and ranges
v. High electrical efficiency

7.5.3 Inductive transducers

Inductive transducers are passive transducers which operate generally one of the following parameters.

i. Variation of self-inductance of the coil
ii. Variation of mutual-inductance of the coil
iii. Production of eddy currents

a. Working principle

The self inductance of a coil is given by,

$$L = \frac{N^2}{l/\mu A} = \frac{N^2 \mu A}{l}$$

where L = self inductance
N = Number of turns
μ = permeability of magnetic materials
A = Area of cross-section of magnetic path
l = lengnth of mean turn

From the above expression it is clear that, the self-inductance of the coil can be varied by varying the number of turns on the coil (N), permeability μ , Area of cross-section (A) and length of mean turn (l). These transducers are usually used for the measurement of linear and rotary displacement. The following figures shows the schematic diagram of linear and rotary type inductive transducers.

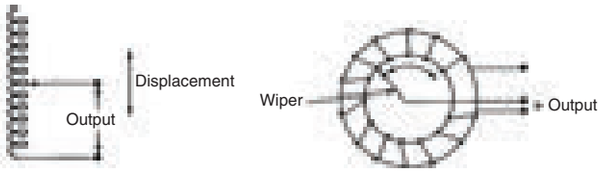


Fig. 7.9 Inductive transducers

b. Linear variable differential transformer (LVDT)

This is most widely used inductive transducer for translating linear movement into electrical signal. The basic construction of LVDT is shown below.

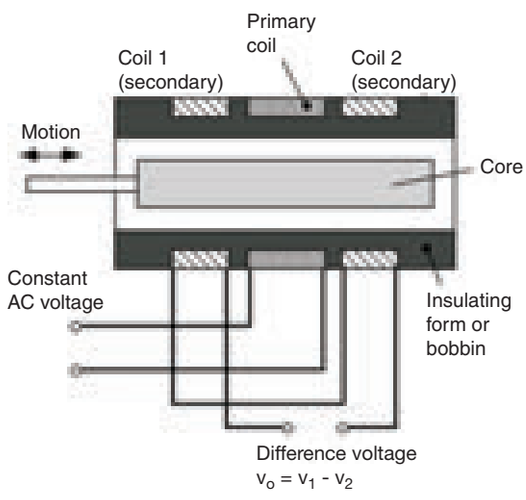


Fig. 7.10 LVD transformer

LVDT is working on the principle of a Transformer only. Like a transformer LVDT has both primary and secondary. Primary winding is connected to an AC supply. The secondary has two identical windings which are connected in series opposition. A soft iron core is attached

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